

packages/graph-view/src/utils/containers.ts

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Overview

`packages/graph-view/src/utils/containers.ts` is a new module that derives logical containers from graph nodes. It groups nodes by folder structure, falls back to community detection when folder grouping is insufficient, and suppresses single-child containers for larger graphs.

Key changes

- **Imports** – `GraphNode / GraphEdge` from `@novadiff/graph-core/types` and `detectCommunities` from `./louvain` (lines 1-5).
- **Interfaces** – `DerivedContainer` (`id, name, nodeIds, strategy`) and `DeriveResult` (`containers, ungrouped`) (lines 7-17).
- **Helper functions**
 - `commonPrefix(paths)` (lines 30-45) finds the longest common directory prefix.
 - `firstSegment(path)` (lines 47-50) extracts the first path segment.
 - `groupByFolder(nodes)` (lines 52-75) groups nodes by the first segment after trimming the common prefix.
 - `shouldFallbackToCommunity(groups, rooted, totalNodes)` (lines 77-89) decides whether to use community detection based on bucket count and concentration thresholds.
- **Main export** – `deriveContainers(nodes, edges)` (lines 91-156) orchestrates grouping, optional community detection, container creation, and suppression of single-child containers when `nodes.length ≥ MIN_NODES_FOR_SUPPRESSION`.

Impact

- **Correctness** – deterministic container derivation; community fallback is used when bucket count < `MIN_BUCKET_COUNT` or any bucket exceeds `MAX_CONCENTRATION`.
- **Maintainability** – all container logic resides in a single module with clear type contracts.
- **Performance** – community detection (`detectCommunities`) is invoked only when needed; suppression is a linear pass over containers.
- **Compatibility** – no existing API changes; new exports are additive.
- **Observability** – exported functions can be unit-tested; thresholds are exposed as constants.

Risks & follow-ups

- **Community detection accuracy** – unknown from the available diff; verify clusters for typical graph shapes.
- **Suppression edge cases** – ensure nodes in single-child containers are moved to `ungrouped` when `nodes.length ≥ MIN_NODES_FOR_SUPPRESSION`.

- **Folder grouping correctness** – test `commonPrefix` and `firstSegment` with paths lacking slashes or containing only a root.
- **Integration** – downstream consumers must handle the new `strategy` field (`folder` vs `community`).